



Tuesday, May 10, 2022

Fibertec Project Number: A08134
Project Identification: NPDES- Semi Annual PFAS /
Submittal Date: 04/21/2022

Mrs. Cindy Tomaszewski
City of Cadillac
200 Lake Street
Cadillac, MI 49601

Dear Mrs. Tomaszewski,

Thank you for selecting Fibertec Environmental Services as your analytical laboratory. The samples you submitted have been analyzed in accordance with NELAC standards and the results compiled in the attached report. Any exceptions to NELAC compliance are noted in the report. These results apply only to those samples submitted. Please note TO-15 samples will be disposed of 7 calendar days after the reporting date. All other samples will be disposed of 30 days after the reporting date.

If you have any questions regarding these results or if we may be of further assistance to you, please contact me at (517) 699-0345.

Sincerely,

A handwritten signature in black ink that reads "Sue Ricketts". The signature is fluid and cursive.

By Sue Ricketts at 10:45 AM, May 10, 2022

For Daryl P. Strandbergh
Laboratory Director

Enclosures

1914 Holloway Drive
11766 E. Grand River
8660 S. Mackinaw Trail

Holt, MI 48842
Brighton, MI 48116
Cadillac, MI 49601

T: (517) 699-0345
T: (810) 220-3300
T: (231) 775-8368

F: (517) 699-0388
F: (810) 220-3311
F: (231) 775-8584

Client Identification:	City of Cadillac	Sample Description:	Field Blank	Chain of Custody:	203803
Client Project Name:	NPDES- Semi Annual PFAS	Sample No:		Collect Date:	04/21/22
Client Project No:	NA	Sample Matrix:	Blank: Field	Collect Time:	11:50

Sample Comments:

Definitions: Q: Qualifier (see definitions at end of report) NA: Not Applicable ‡: Parameter not included in NELAC Scope of Analysis.

PFAS						Aliquot ID:	A08134-001	Matrix: Blank: Field		
Method: EPA 0537.1 (Modified)						Description:	Field Blank			
Parameter(s)	Result	Q	Units	Reporting Limit	Dilution	Preparation		Analysis		
						P. Date	P. Batch	A. Date	A. Batch	Init.
‡ 1. ADONA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 2. 9CI-PF3ONS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 3. 11CI-PF3OUdS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 4. N-EtFOSAA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 5. FtS 4:2	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 6. FtS 6:2	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 7. FtS 8:2	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 8. HFPO-DA	U	C+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 9. N-MeFOSAA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 10. PFBA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 11. PFBS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 12. PFDA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 13. PFDoA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 14. PFDS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 15. PFHpA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 16. PFHpS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 17. PFHxA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 18. PFHxS-Total	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 19. PFNA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 20. PFNS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 21. PFOA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 22. PFOSA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 23. PFOS-Total	3.0	C+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 24. PFPeA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 25. PFPeS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 26. PFTeA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 27. PFTrIA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 28. PFUnA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG

Client Identification:	City of Cadillac	Sample Description:	Effluent	Chain of Custody:	203803
Client Project Name:	NPDES- Semi Annual PFAS	Sample No:		Collect Date:	04/21/22
Client Project No:	NA	Sample Matrix:	Wastewater	Collect Time:	11:55

Sample Comments:

Definitions: Q: Qualifier (see definitions at end of report) NA: Not Applicable ‡: Parameter not included in NELAC Scope of Analysis.

PFAS						Aliquot ID:	A08134-002	Matrix: Wastewater			
Method: EPA 0537.1 (Modified)						Description: Effluent					
Parameter(s)	Result	Q	Units	Reporting Limit	Dilution	Preparation		Analysis			
						P. Date	P. Batch	A. Date	A. Batch	Init.	
‡ 1. ADONA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 2. 9CI-PF3ONS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 3. 11CI-PF3OUdS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 4. N-EtFOSAA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 5. FtS 4:2	U	EIS+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 6. FtS 6:2	U	EIS+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 7. FtS 8:2	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 8. HFPO-DA	U	C+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 9. N-MeFOSAA	1.0		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 10. PFBA	4.6		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 11. PFBS	32		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 12. PFDA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 13. PFDoA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 14. PFDS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 15. PFHpA	2.0		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 16. PFHpS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 17. PFHxA	23		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 18. PFHxS-Total	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 19. PFNA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 20. PFNS	U	F+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 21. PFOA	2.0		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 22. PFOSA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 23. PFOS-Total	3.1	C+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 24. PFPeA	4.5		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 25. PFPeS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 26. PFTeA	U	EIS-	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 27. PFTrIA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	
‡ 28. PFUnA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG	

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F: (810) 220-3311
F: (231) 775-8584

Client Identification:	City of Cadillac	Sample Description:	Effluent Duplicate	Chain of Custody:	203803
Client Project Name:	NPDES- Semi Annual PFAS	Sample No:		Collect Date:	04/21/22
Client Project No:	NA	Sample Matrix:	Wastewater	Collect Time:	11:58

Sample Comments:

Definitions: Q: Qualifier (see definitions at end of report) NA: Not Applicable ‡: Parameter not included in NELAC Scope of Analysis.

PFAS						Aliquot ID:	A08134-003	Matrix: Wastewater		
Method: EPA 0537.1 (Modified)						Description:	Effluent Duplicate			
Parameter(s)	Result	Q	Units	Reporting Limit	Dilution	Preparation		Analysis		
						P. Date	P. Batch	A. Date	A. Batch	Init.
‡ 1. ADONA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 2. 9CI-PF3ONS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 3. 11CI-PF3OUdS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 4. N-EtFOSAA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 5. FtS 4:2	U	EIS+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 6. FtS 6:2	U	EIS+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 7. FtS 8:2	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 8. HFPO-DA	U	C+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 9. N-MeFOSAA	1.0		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 10. PFBA	5.2		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 11. PFBS	30		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 12. PFDA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 13. PFDoA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 14. PFDS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 15. PFHpA	2.1		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 16. PFHpS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 17. PFHxA	23		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 18. PFHxS-Total	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 19. PFNA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 20. PFNS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 21. PFOA	1.9		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 22. PFOSA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 23. PFOS-Total	7.2	C+	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 24. PFPeA	4.6		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 25. PFPeS	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 26. PFTeA	U	EIS-	ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 27. PFTrIA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG
‡ 28. PFUnA	U		ng/L	1.0	1.0	05/02/22	PS22E02F	05/02/22	SM22E02A	SKG

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F: (231) 775-8584

Acronym (Param)	Analyte Name	CAS Number
1. ADONA	4,8-dioxa-3H-perfluorononanoic acid	919005-14-4
2. 9Cl-PF3ONS	9-chlorohexadecafluoro-3-oxanone-1-sulfonic acid	756426-58-1
3. 11Cl-PF3OUdS	11-chloroeicosafluoro-3-oxaundecane-1-sulfonic acid	763051-92-9
4. N-EtFOSAA	2-(N-Ethylperfluorooctanesulfonamido) acetic acid	2991-50-6
5. FtS 4:2	Fluorotelomer sulphonic acid 4:2	757124-72-4
6. FtS 6:2	Fluorotelomer sulphonic acid 6:2	27619-97-2
7. FtS 8:2	Fluorotelomer sulphonic acid 8:2	39108-34-4
8. HFPO-DA	Hexafluoropropylene oxide dimer acid	13252-13-6
9. N-MeFOSAA	2-(N-Methylperfluorooctanesulfonamido) acetic acid	2355-31-9
10. PFBA	Perfluorobutanoic acid	375-22-4
11. PFBS	Perfluorobutanesulfonic acid	375-73-5
12. PFDA	Perfluorodecanoic acid	335-76-2
13. PFDoA	Perfluorododecanoic acid	307-55-1
14. PFDS	Perfluorodecanesulfonic acid	335-77-3
15. PFHpA	Perfluoroheptanoic acid	375-85-9
16. PFHpS	Perfluoroheptanesulfonic acid	375-92-8
17. PFHxA	Perfluorohexanoic acid	307-24-4
18. PFHxS-Total	Perfluorohexanesulfonic acid	355-46-4
19. PFNA	Perfluorononanoic acid	375-95-1
20. PFNS	Perfluoronananesulfonic acid	68259-12-1
21. PFOA	Perfluorooctanoic acid	335-67-1
22. PFOSA	Perfluorooctanesulfonamide	754-91-6
23. PFOS-Total	Perfluorooctanesulfonic acid	1763-23-1
24. PFPeA	Perfluoropentanoic acid	2706-90-3
25. PFPeS	Perfluoropentanesulfonic acid	2706-91-4
26. PFTeA	Perfluorotetradecanoic acid	376-06-7
27. PFTriA	Perfluorotridecanoic acid	72629-94-8
28. PFUnA	Perfluoroundecanoic acid	2058-94-8

Definitions/ Qualifiers:

- A:** Spike recovery or precision unusable due to dilution.
B: The analyte was detected in the associated method blank.
E: The analyte was detected at a concentration greater than the calibration range, therefore the result is estimated.
J: The concentration is an estimated value.
M: Modified Method
U: The analyte was not detected at or above the reporting limit.
X: Matrix Interference has resulted in a raised reporting limit or distorted result.
W: Results reported on a wet-weight basis.
***:** Value reported is outside QC limits

Exception Summary:

- C+** : Recovery in the Reporting Limit Check Sample (RLCS) exceeds the upper control limit. Results may be biased high.
EIS- : The Isotope Dilution/Extracted Internal Standard area exceeds the lower control limit.
EIS+ : The Isotope Dilution/Extracted Internal Standard area exceeds the upper control limit.
F+ : Recovery from the spiked aliquot exceeds the upper control limit (matrix spike or matrix spike duplicate).

Analysis Locations:

All analyses performed in Holt.



Accreditation Number(s):

T104704518-19-8 (TX)

1914 Holloway Drive
11766 E. Grand River
8660 S. Mackinaw Trail

Holt, MI 48842
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Cadillac, MI 49601

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F: (810) 220-3311
F: (231) 775-8584

Client Name: <u>City of Cadillac - WWTP</u>			PARAMETERS										Matrix Code		Deliverables												
Contact Person: <u>Cindy Tomaszewski</u>			MATRIX (SEE RIGHT CORNER FOR CODE)	# OF CONTAINERS	<u>PFAS (MI 28 List)</u>									S Soil	GW Ground Water		Level 2										
Project Name/ Number: <u>NPDES - SEMI ANNUAL PFAS</u>																		HOLD SAMPLE	A Air	SW Surface Water	Level 3						
Email distribution list: <u>ctomaszewski OR lab@cadillac-mi.net</u>																						O Oil	WW Waste Water	Level 4			
Quote#																									P Wipe	x Other: Specify	☒ EDD
Purchase Order# <u>to be determined (requested)</u>																											
Date	Time	Sample #	Client Sample Descriptor																								
<u>4/19/22</u>	<u>08:00</u>	<u>11580</u>	<u>Field Blank</u>	<u>X</u>	<u>1</u>	<u>X</u>																					
<u>4/21/22</u>	<u>1155</u>	<u>1155</u>	<u>Duplicate Effluent</u>	<u>ww</u>	<u>3</u>	<u>X</u>																					
<u>4/21/22</u>	<u>1158</u>	<u>1158</u>	<u>Effluent Duplicate</u>	<u>ww</u>	<u>3</u>	<u>X</u>																					
Comments: <u>Any questions - Just call. Thx CAT</u>																											
Sampled/Relinquished By: <u>C. Tomaszewski</u>			Date/Time: <u>4/21/2022 @ 1400</u>			Received By: <u>P. Schweik Guswald</u> <u>on ice</u>																					
Relinquished By: <u>P. Schweik Guswald</u>			Date/Time: <u>4-22-22 1430</u>			Received By: <u>Andrew</u>																					
Relinquished By: <u>Andrew</u>			Date/Time: <u>4/22/22 16:55</u>			Received By Laboratory: <u>Manz Power</u>																					
Turnaround Time ALL RESULTS WILL BE SENT BY THE END OF THE BUSINESS DAY ☒ 1 bus. day ☐ 2 bus. days ☐ 3 bus. days ☐ 4 bus. days ☒ 10-14 bus. days (standard) <u>for PFAS</u> Other (specify time/date requirement): _____																											
Fibertec project number: <u>A08134</u> Temperature upon receipt at Lab: <u>5.3°C</u> Received On Ice																											
Please see back for terms and conditions																											